

Enhancing customer retention with machine learning: A comparative analysis of ensemble models for accurate churn prediction

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ABSTRACT

This paper investigates the use of machine learning models for customer churn prediction, focusing on the comparative effectiveness of ensemble approaches such as XGBoost and Random Forest with classical classifiers. The study evaluates the benefits and shortcomings of each strategy in dealing with complicated datasets by analyzing confusion matrices and Receiver Operating Characteristic (ROC) curves in detail. Ensemble models outperformed on key criteria such as accuracy, precision, recall, and F1 scores, yielding excellent results. These results demonstrate the effectiveness of ensemble approaches in producing accurate and trustworthy forecasts, making them suitable for client retention efforts. The report offers practical insights for firms looking to use sophisticated machine learning approaches to make better strategic decisions and retain more customers.

1. Introduction

In today's highly competitive business environment, understanding customer behavior has become essential for organizations seeking to enhance their decision-making and maintain a competitive edge (Onari et al., 2024). The rise of digital technologies and data analytics has transformed the way businesses gather and interpret customer information, enabling a shift from intuition-based decisions to data-driven strategies. By analyzing various data sources such as transaction histories, behavioral patterns, and customer feedback, companies can gain deeper insights into customer preferences and behaviors. This shift allows for more personalized marketing, optimized product offerings, and improved customer retention, ultimately driving business growth and enhancing customer experiences. Understanding customer behavior is crucial for businesses striving to maintain a competitive edge and ensure long-term success. A critical aspect of customer behavior is customer churn, which refers to the rate at which customers discontinue their relationship with a company or service provider (Awan et al., 2021; Bousdekis et al., 2021; Calza et al., 2023). Analyzing customer churn helps businesses identify the key reasons behind customer attrition, enabling them to devise strategies to improve customer retention and satisfaction. This data-driven approach effectively transforms insights into customer behavior into actionable business strategies (Khodabandehlou and Zivari Rahman, 2017). The integration of

sophisticated data analytics and machine learning has moved decision-making processes away from intuition-based tactics and toward data-driven approaches (Anjorin et al., 2024). This paradigm change allows businesses to derive actionable insights from a variety of data sources, including transaction histories, behavioral patterns, and feedback, resulting in targeted marketing, optimized product offers, and improved retention methods. Customer churn is a crucial concern in this industry, since it has a considerable influence on firms, particularly those that rely on subscription-based revenue models (JUBI, 2024).

Customer churn represents a significant challenge for many businesses, particularly those that rely on subscription-based or recurring revenue models. A high churn rate can lead to reduced revenue, increased marketing costs to acquire new customers, and potential damage to the company's reputation (Afzal et al., 2024). By understanding churn, businesses can identify the factors contributing to customer dissatisfaction and attrition. This understanding facilitates the development of targeted interventions to improve the customer experience, offer personalized incentives, and address specific pain points (Janzer, 2020). Predictive analytics plays a crucial role in the context of customer churn. By analyzing historical customer data and identifying patterns associated with churn, businesses can develop predictive models to forecast which customers are at risk of leaving. This proactive approach enables companies to implement preventative measures, such as personalized retention campaigns, enhanced customer support, or

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customized offers, thereby reducing churn risk. Machine learning algorithms, such as logistic regression, random forests, and gradient boosting, allow for accurate predictions and deeper insights into customer behavior (Singh et al., 2024). Data-driven customer churn analysis provides valuable insights into customer behavior, enabling businesses to make informed strategic decisions. By integrating churn predictions with customer segmentation and behavioral data, companies can prioritize their efforts on high-risk segments, optimize resource allocation, and design targeted retention strategies. For example, identifying customers who exhibit dissatisfaction or reduced engagement allows businesses to intervene with personalized offers or enhanced service levels. This approach helps retain valuable customers and improves customer satisfaction and loyalty (Manzoor et al., 2024). Understanding the factors contributing to customer churn empowers businesses to enhance their overall customer experience. Data-driven insights help companies identify common characteristics or behaviors among churn-prone customers (Karwa, 2025), such as service usage patterns, support interactions, or payment issues. Addressing these factors through improved service quality, personalized communication, and proactive support can significantly reduce churn rates.

Customer churn analysis not only identifies the reasons behind customer attrition but also equips organizations with predictive tools to mitigate it (Hossain et al., 2020). Predictive analytics, powered by machine learning, provides proactive mechanisms for addressing churn risks. By analyzing historical data and employing models such as logistic regression, Random Forest, and XGBoost, businesses can forecast churn tendencies and implement targeted retention strategies. These strategies include personalized offers, enhanced service levels, and proactive customer support. This research contributes to the field by systematically evaluating machine learning models for their efficacy in churn prediction. The novelty lies in the detailed comparison of traditional classifiers and ensemble methods through confusion matrices and ROC curve analysis. By highlighting the superior performance of ensemble approaches like XGBoost and Random Forest, the study provides a robust framework for businesses seeking reliable predictive tools for customer retention (Rosário and Dias, 2023).

Additionally, businesses can leverage customer feedback and satisfaction surveys to continuously refine their strategies and adapt to evolving customer needs (Vemulapalli, 2024). Uncovering customer behavior through data-driven analysis is essential for strategic decision-making in modern businesses. By focusing on customer churn and utilizing predictive analytics, companies can gain actionable insights into customer behavior, implement targeted retention strategies, and enhance overall customer satisfaction. This proactive approach helps retain valuable customers and strengthens a more customer-centric and resilient business model.

The major objective of this research is to assess the efficacy of several machine learning models, with specific emphasis on ensemble approaches, in precisely forecasting customer turnover. The research's originality is in its comprehensive evaluation of model performance using confusion matrix analysis and Receiver Operating Characteristic (ROC) curves, offering a nuanced knowledge of how various models manage intricate datasets. This work enhances the field by identifying ensemble approaches, namely XGBoost and Random Forest, as optimal methods for churn prediction via a comprehensive analysis of key performance measures, including accuracy, precision, recall, and F1 score. The study underscores the need of balancing model sensitivity with specificity, providing insights for reducing categorization mistakes. These contributions seek to assist organizations in identifying the most effective and dependable models for client retention tactics, facilitating data-driven decision-making in competitive market contexts. This study improves our knowledge of machine learning in customer churn prediction by:

- Showing the relative efficacy of several machine learning models.

- Providing a nuanced assessment of model performance utilizing important measures including as accuracy, precision, recall, and F1 scores.
- Introducing ensemble techniques as improved tools for predicting accuracy in complicated datasets.

Providing useful insights for incorporating predictive analytics into strategic decision-making frameworks.

The objective of this research study is to fill in the gaps in the existing knowledge base with regard to the relative performance level of various machine learning ensembling methods, including XGBoost and Random Forest, combined with traditional models such as logistic regression and KNN for customer churn prediction. This study further attempts to identify the most influencing factors that cause customer churn and tests the effectiveness of ensemble models in managing class imbalance with a view to reducing classification errors.

The hypotheses for these purposes are that, for the stated goals, the ensemble methods, XGBoost and Random Forest, would considerably outperform traditional classifiers in terms of accuracy, precision, recall, and F1 scores for churn prediction. Also, XGBoost, due to its superior handling of complex data patterns and imbalances, is expected to show the highest magnitude of performance among the models tested. Finally, feature importance analyses are expected to identify tenure, contract type, and monthly charges as some of the most important predictors of customer churning.

This research used a telecommunications dataset of 10,000 customer records from a telecom company, encompassing demographics, subscription details, service usage patterns, payment methods, and churn indicators. Essential characteristics include customer ID, age, gender, duration (in months), contract type, monthly costs, total charges, type of internet service, type of phone service, and a binary churn indicator (1 for churn, 0 for non-churn). The data was acquired over an unspecified period but underwent extensive preparation to ensure quality. Missing values in numerical columns, such as total charges, were imputed using median values, while missing categorical values were substituted with the most frequent category. Categorical variables were encoded using one-hot encoding for multi-class variables and binary encoding for binary-class variables. Numerical features were normalized using Min-Max normalization to provide comparability across varying ranges. The dataset exhibited a class imbalance (20 % churn, 80 % non-churn), which was rectified by the use of the Synthetic Minority Oversampling Technique (SMOTE) only on the training set to prevent data leakage. The dataset was divided into training (70 %), validation (15 %), and testing (15 %) subsets. The dataset underwent further scrutiny using exploratory data analysis (EDA), interquartile range (IQR) analysis for outlier detection, and nearest-neighbor imputation for extreme values. Feature selection used correlation analysis and Recursive Feature Elimination (RFE) to choose the most significant predictors. The assessed models included logistic regression, K-Nearest Neighbors (KNN), Random Forest, Gradient Boosting, and XGBoost, with Random Forest and XGBoost outperforming the others in accuracy, precision, recall, and F1 scores. The study found tenure, contract type, and monthly expenses as the major factors in churn prediction. Despite exceptional performance, possible biases emerge from synthetic data generation (SMOTE), the risk of overfitting in ensemble methods, and challenges in adapting models to changing consumer behavior. Incremental learning, drift detection methodologies, and cost-sensitive learning were suggested as potential research directions to address these challenges. The dataset's validity was further confirmed using 5-fold cross-validation and comprehensive evaluation metrics, including accuracy, precision, recall, F1-score, ROC AUC, MCC, and log loss, so ensuring a thorough assessment of model performance.

The rest of this paper is organized as follows. Section 2 examines the relevant literature on customer behavior and churn analysis. Section 3 describes the approach, including data sources and analysis methodologies. Section 4 describes the models used and their performance

indicators. Section 5 discusses the data and compares model effectiveness. The findings are discussed comprehensively in the sixth section. Besides, cost implications and computing resource considerations have been given in the seventh section. Section 8 closes by discussing the study's implications, limitations, and future research objectives.

2. Literature review

The study of customer behavior has been a central focus in marketing and business strategy for decades, with numerous theories and models developed to explain how and why consumers make purchasing decisions (Campi et al., 2021; Chen et al., 2021; Gutierrez-Franco et al., 2021; Hosen et al., 2024). Early research primarily relied on psychological and sociological frameworks, such as Maslow's hierarchy of needs and the theory of planned behavior, to understand the motivations behind consumer choices. However, with the rise of big data and advanced analytics, contemporary studies have shifted towards data-driven approaches that integrate quantitative analysis with traditional behavioral models. Scholars have increasingly focused on leveraging tools such as machine learning, predictive analytics, and customer segmentation to gain more accurate and granular insights into consumer behavior patterns (Chen et al., 2012). This evolving landscape has opened new avenues for understanding customer preferences, loyalty drivers, and decision-making processes, providing businesses with actionable insights for strategic decision-making (Johnson et al., 2021). Onari et al. (2024) proposed a decision support framework that addressed challenges in credit scoring by enhancing algorithm transparency and guiding rejected customers. They utilized Shapley additive explanations and a game-based model to generate successful counterfactual predictions for improving customer applications. Rawat et al. (2024) assessed the simplicity and effectiveness of neural networks compared to more traditional methods, demonstrating that their approach, which included network pruning and interpretability-based regularization, proved more accurate, faster, and easier to understand. They showed that small neural networks could simplify models without compromising accuracy, making them more accessible for practical use.

Furthermore, a study conducted by Zadeh et al. (2011) attempted to categorize bank clients according to their behaviors to assist the bank in retaining customers and acquiring new ones. Awan et al. (2021) examined the critical role of Big Data Analytics (BDA) and Business Intelligence & Analytics (BI&A) in enhancing decision-making quality within organizations, particularly in the context of supporting the circular economy (CE). By empirically investigating Czech manufacturing firms, the research demonstrates that while BDA capability directly improved decision-making, data-driven insights do not mediate this relationship, but are essential for BI&A to positively influence decision-making. These findings offered valuable guidance for managers aiming to leverage BDA and data-driven insights to drive CE performance. Troisi et al. (2020) examined the impact of Big Data analytics and cognitive computing, via the Growth Hacking model, on marketing decision-making in the B2B sector by employing a data-driven methodology, utilizing Action Research to identify strategies and tactics that improve customer relationships, foster innovation, and facilitate ongoing product development (Troisi et al., 2020; Sala et al., 2022; Zaitsava et al., 2022).

Moreover, Sajwan and Tripathi (2024) utilized Market Basket Analysis, employing algorithms like Apriori, to uncover hidden associations between products in consumer purchase data, offering businesses strategic insights into evolving customer preferences and behavior, thereby helping them make informed decisions to stay competitive in dynamic markets. Nadler and McGuigan (2018) demonstrated that advertisers and marketers using concepts and methodologies from behavioral economics convey a markedly different narrative to their customers on the objectives and use of digital marketing and consumer surveillance. This discussion revealed that some digital marketers see their tactics as mechanisms of social control, using principles from

behavioral economics to discern customers' cognitive and emotional biases and exploit their weaknesses. Zaitsava et al. (2022) investigated how intuitive and rational reasoning biases interact in data-driven decision-making (DDDM), uncovering cognitive and data biases through ethnographic research at a UK transportation firm, while highlighting the role of trust in data and proposing managerial implications. Eslami et al. (2024) explored customer preferences in the IoT context through segmentation using the self-organizing map (SOM) algorithm and a decision tree model, uncovering key factors impacting customer satisfaction, retention, and loyalty, offering strategic insights for businesses to optimize IoT-CRM strategies. Hossain et al. (2023) examined how sustained application of customer analytics enhances market effectiveness in retail, using a model grounded in market orientation theory, and highlights the role of analytics in boosting operational performance in a big data environment.

Recent studies conducted by Mach-Król and Hadasik (2021), Hassouna et al. (2016) aimed to examine data mining techniques, specifically decision trees and logistic regression, in the development of a customer churn model. The study utilized a real-world dataset obtained from an English mobile provider. The dataset consisted of two balanced subsets: one for training, which included 17 variables and 19,919 customers, and another for testing, which included 17 variables and 15,519 consumers. The authors employed multiple decision tree algorithms, such as CART, C5.0, and CHAID, to compare them with logistic regression. The evaluation criteria consisted of the Area Under the Curve (AUC) of the Receiver Operating Characteristic (ROC) curve, the performance of the top decile, and the overall accuracy. Remarkably, the C5.0 decision tree algorithm exhibited superior performance to other algorithms, such as logistic regression, with an AUC of 0.763. This conclusion contradicts previous research (Owczarczuk, 2010).

It should be noted that Nilashi et al. (2021) presented a new machine-learning method for evaluating Sustainability, using self-organizing map (SOM) clustering and classification and regression trees (CART). They showcased the robust predictive accuracy of CART models and underscored the efficacy of ensemble approaches. Using the fuzzy evaluation (SAFE) dataset, they evaluated the proposed technique for sustainability assessment, demonstrating its potential as an automated decision-making tool and providing valuable insights into addressing considerable data challenges with machine learning. Troisi et al. (Troisi and Maione, 2024) explored the significance of data-driven decision-making in empowering businesses to make informed decisions swiftly and effectively. By leveraging advanced analytics techniques and machine learning algorithms, the researchers highlighted how organizations can uncover valuable patterns, automate tasks, and optimize operations, demonstrating the transformative impact of data-driven approaches across various industries. Govindarajan et al. (Govindarajan and Ananthanpillai, 2024) explored the critical role of data analytics in transforming product decision-making and enhancing processes from concept to market launch. The researchers used advanced data analysis techniques to demonstrate how companies can gain valuable insights, optimize product designs, and develop targeted marketing strategies while addressing data integration, quality, and security challenges. Case studies illustrated the significant benefits of data-driven approaches in product development, emphasizing the need to address ethical and practical obstacles to realize their potential fully. Also more studies have been reviewed in Table 1 to highlight the major gaps and shortcomings of the previous works.

This research identifies some significant deficiencies in the current literature about customer turnover prediction. Although several studies have investigated different machine learning models for churn prediction, there is a paucity of research providing a thorough comparison between conventional approaches like K-Nearest Neighbors and Logistic Regression and sophisticated ensemble models such as XGBoost and Random Forest. This work addresses the gap by delivering a comprehensive performance assessment across many measures, emphasizing the enhanced accuracy and dependability of ensemble methods. This

Table 1
A brief review of the related works based on the main methods and aims.

No.	Refs.	Methods Used	Focus/Objective	Key Findings
1	Onari et al. (2024)	Decision Trees, Logistic Regression	Developing explainable credit scoring frameworks	SHAP improved interpretability and guided rejected customers
2	Awan et al., 2021)	Big Data Analytics (BDA), BI&A	Enhancing decision-making for the circular economy	BDA improved decision-making in manufacturing firms
3	Rawat et al. (2024)	Neural Networks with Regularization	Simplifying neural networks for practical use	Network pruning retained accuracy while enhancing usability
4	Zadeh et al. (2011)	Clustering (SOM), CART	Segmenting bank clients to improve retention	Identified key customer groups for retention and acquisition
5	Troisi et al. (2020)	Big Data Analytics, Action Research	Improving marketing decisions in the B2B sector	Highlighted the role of Growth Hacking in fostering innovation
6	Sala et al. (2022)	Growth Hacking, Data-Driven Marketing	Enhancing customer relationships and innovation	Demonstrated cognitive computing's role in decision-making
7	Zaitsava et al. (2022)	Ethnographic Research	Exploring biases in data-driven decision-making	Identified cognitive and data biases in organizational decisions
8	Sajwan and Tripathi (2024)	Market Basket Analysis (Apriori Algorithm)	Identifying associations in purchase data	Offered insights into evolving customer preferences
9	Nadler and McGuigan (2018)	Behavioral Economics	Understanding cognitive and emotional biases	Revealed social control tactics in digital marketing
10	Eslami et al. (2024)	Clustering (SOM), Decision Tree Model	Segmenting IoT customers	Identified factors impacting satisfaction, retention, and loyalty
11	Hossain et al. (2023)	Market Orientation Theory	Enhancing retail market effectiveness with analytics	Demonstrated analytics' role in operational performance
12	Mach-Król and Hadasik (2021)	Decision Trees (CART, C5.0, CHAID), Logistic Regression	Developing churn prediction models	C5.0 outperformed logistic regression with higher AUC (0.763)
13	Hassouna et al. (2016)	Decision Trees, Random Forest	Churn prediction in the telecom sector	Showed Random Forest's ability to manage large telecom datasets
14	Owczarczuk (2010)	Logistic Regression, SVM	Comparing traditional churn prediction models	Highlighted the limitations of logistic regression in churn
15	Nilashi et al. (2021)	SOM Clustering, CART	Evaluating sustainability with machine learning	Showcased CART's predictive accuracy and ensemble methods' potential
16	Troisi and Maione (2024)	Advanced Analytics,	Empowering businesses with	Highlighted transformative

Table 1 (continued)				
No.	Refs.	Methods Used	Focus/Objective	Key Findings
17	Govindarajan and Ananthanpillai (2024)	Machine Learning Advanced Data Analysis Techniques	data-driven decisions Enhancing product decision-making from concept to market	potential of data-driven strategies Provided insights into optimizing product designs and strategies

paper presents a novel comparative analysis of traditional and ensemble machine learning models, emphasizing the superior performance of ensemble methods like XGBoost and Random Forest in accurately predicting customer churn using detailed evaluation metrics such as confusion matrices and ROC curves.

3. Methodology

This research employs a data-driven technique to analyze consumer behavior, using both qualitative and quantitative data from diverse client encounters, such as transaction histories, website analytics, and customer feedback. Essential indicators, including Customer Lifetime Value (CLV), churn rate, and purchase frequency, are monitored and examined to discern trends in buying behavior and customer preferences. Advanced segmentation techniques—demographic, behavioral, and psychographic—are used to classify clients into specific categories for enhanced analysis. Predictive analytics technologies, including machine learning models, are used to anticipate future patterns, including churn risk and product demand. The research includes case studies that exemplify effective data-driven methods used by prominent corporations. Integrating data from many sources and using real-time analytic tools are essential for obtaining actionable insights for strategic decision-making.

The experimental technique began with data preparation, which included missing value management, categorical category encoding, Min-Max normalization of numerical features, and dataset balancing using SMote for training data. The dataset was divided into three sets: 70 % training, 15 % validation, and 15 % testing. Model training included logistic regression, K-Nearest Neighbors (KNN), Random Forest, Gradient Boosting, and XGBoost, all of which were improved using grid search and 5-fold cross-validation to fine-tune hyperparameters. Logistic regression with a tailored regularizing strength ($C \in \{0.01, 0.1, 1, 10\}$) using L2 regularization (Ridge). KNN was adjusted for Euclidean, Manhattan distance metric, and neighbor count ($k = 3, 5, 7, 9$). Random Forest was changed based on the number of trees ($n_estimators \in \{100, 300, 500\}$, maximum depth (3–20), and minimum sample size per split (2–10). Gradient Boosting (XGBoost) was optimized for the learning rate (0.01–0.3), maximum depth (3–10), number of boosting rounds (50–500), and L1/L2 regularizing parameters. Model evaluation ensured resistance against skewed data by using accuracy, precision, recall, F1-score, ROC AUC, MCC, and log loss metrics. To assess significance in developments, paired *t*-tests and Wilcoxon signed-rank tests were used to statistically compare model performance. The statistical validity of ensemble models' superiority was established by an analysis of classification error changes using McNemar's test.

3.1. Preprocessing and dataset

This research employs a dataset of 10,000 records from a telecommunications company, which includes customer demographics, subscription details, service usage patterns, payment methods, and churn indicators. Customer ID, age, gender, tenure (number of months subscribed), contract type, monthly charges, total charges, and a binary churn designation (1 for churn, 0 for non-churn) are among the most significant features. In order to improve the analysis, additional information regarding the type of internet service, phone service, and

payment mechanisms was supplied.

The initial step in data preprocessing was to sanitize the data to remove anomalies, duplicates, and absent values. The median values were used to impute missing values in numerical columns, such as total charges, while the most prevalent category was used to populate categorical columns with missing entries. For multi-class variables (e.g., internet service type), categorical features were encoded using one-hot encoding, while binary encoding was employed for two-class variables (e.g., gender). In order to guarantee comparability across diverse ranges, numerical features were scaled using Min-Max normalization. The class imbalance in the target variable, churn, was substantial (20 % churn vs. 80 % non-churn). To resolve this issue, the training set was subjected to the Synthetic Minority Oversampling Technique (SMOTE) in order to effectively balance the classes. In order to guarantee the robustness of the model's training and evaluation, the data was segmented into three subsets: training (70 %), validation (15 %), and testing (15 %).

While this research emphasizes the better effectiveness of ensemble approaches such as XGBoost and Random Forest in predicting customer turnover, it does not address the models' shortcomings in dealing with unbalanced data or fluctuating dataset sizes. Imbalanced datasets, with significantly fewer churn instances than non-churn instances, can result in biased predictions for the majority class. Although strategies such as the Synthetic Minority Oversampling Technique (SMOTE) were utilized to alleviate this, ensemble models may still be sensitive to class imbalance, particularly when default hyperparameters are used. Furthermore, shifting dataset sizes may influence model performance, as bigger datasets may improve learning while smaller datasets may result in overfitting or under fitting, especially in complicated models like XGBoost. Logistic regression, although less accurate overall, is often more resilient to small datasets owing to its simplicity, indicating a trade-off between complexity and stability under changing data circumstances. Future research might look at adaptive approaches for class balance and hyper parameter adjustment to overcome these restrictions more fully.

The dataset was split chronologically to emulate real-world churn prediction scenarios when future data must be expected based on past observations, hence including temporal validation. The data was divided into three categories: training (prior times), validation (mid-range periods), and testing (most recent periods), therefore ensuring that models learned from past customer behavior and were tested against future trends rather than haphazard divisions. In order to replicate real-world deployment, models were continually trained on expanding historical data and assessed on the following time period using a rolling-window cross-valuation method. Performance was tracked over several time periods and adaptive retraining was started when ROC AUC or F1-score dropped by more than a predefined level (e.g., 5 %). This helped to explain idea drift. This method increases real-world applicability and ensures robustness to changing churn patterns.

In predictive modeling, the quality of data is paramount, as missing and noisy data can significantly impact the performance of machine learning models. This study employed several preprocessing techniques to address these challenges and ensure the reliability of the dataset.

3.1.1. Handling missing data

Imputation of missing values was done using imputation methodologies suitable for the type of missing values. The numerical features like "total charges" were imputed with their median value in respective columns, since the median is not that susceptible to outliers as compared to the mean. While for categorical attributes, missing values were filled in with the mode, or most frequent category, so that consistency in the classification would be maintained. These approaches were followed to minimize the bias and keep the integrity of the data. Exploratory data analysis was conducted following preprocessing to make sure that imputation was effective in not leaving any systematic pattern of missingness.

3.1.2. Handling noisy data

IQR analysis was conducted to detect noisy data-for example, outliers or irregular values-for the numeric features. The values falling beyond 1.5 times of IQR limits were marked as potential outliers: for example, extremely high feature values for "monthly charges" were checked and cleaned. All those outliers where the source was traced out to erroneous entry of data against ground truth have been either rectified if that could be found; otherwise, such values were replaced with nearest-neighbor imputation. Inconsistencies like typos or unexpected categories of categorical variables were rectified by cross-validation with domain knowledge or known patterns in the data. For example, an invalid category for contract type was changed to the most probable category, as dictated by the distribution of that data.

3.1.3. Noise reduction in feature scaling

Therefore, Min-Max normalization was applied to scale numerical features, which prevented noisy data from influencing model training. This transformation brought all features into the range between 0 and 1 and reduced the influence of extreme values on distance-based models, including KNN, thus keeping the model consistent.

3.1.4. Impact on model development

These steps were the most important preprocessing for a powerful dataset in terms of real customer behavior. This resolves the problem of missing values and noise within the dataset, ensuring minimum biased or unreliable performance of the model. What is more, such preprocessing techniques were assessed for their impact on the distribution of data and model performance at training to ensure imputation and cleaning did not introduce unwanted distortions.

3.2. Model development

Five machine learning models were constructed and validated for attrition prediction, spanning from classical approaches to advanced ensemble methods. In order to evaluate linear relationships between predictors and the attrition label, logistic regression was implemented as a baseline. K-Nearest Neighbors (KNN) offered a distance-based, non-linear model that was optimized for optimal k-values. In order to improve the predictive accuracy and robustness, ensemble methods, such as Random Forest and Gradient Boosting (e.g., XGBoost), were implemented. These methods were selected for their capacity to mitigate overfitting through aggregation and manage intricate patterns in large datasets. In order to identify the most influential predictors, correlation analysis and Recursive Feature Elimination (RFE) were implemented for feature selection. In order to enhance efficacy and optimize hyperparameters, grid search was implemented for each model.

All models were implemented in Python using Scikit-learn, XGBoost, and NumPy, ensuring reproducibility. Logistic Regression was trained with L2 regularization ($C = 1.0$, solver = 'lbfgs') as a baseline. K-Nearest Neighbors (KNN) used $k = 5$, Euclidean distance, and uniform weighting. Random Forest was optimized with $n_estimators = 500$, $max_depth = 10$, $min_samples_split = 4$. Gradient Boosting (XGBoost) used $learning_rate = 0.1$, $max_depth = 6$, $n_estimators = 300$, $colsample_bytree = 0.8$, $subsample = 0.8$. Hyperparameters were tuned via grid search with 5-fold cross-validation, and evaluation metrics included accuracy, precision, recall, F1-score, ROC AUC, MCC, and log loss to ensure a rigorous comparison.

3.3. Pipeline for machine learning

The machine learning pipeline was developed to integrate the most effective methods for feature engineering, model development, and optimization. The most significant predictors of attrition were selected using recursive feature elimination (RFE) with logistic regression as a baseline model. This step preserved critical predictors such as tenure, contract type, and monthly charges, while simultaneously reducing

computational overhead and noise.

Five machine learning algorithms were employed to process the selected features: logistic regression, K-Nearest Neighbors (KNN), Random Forest, Gradient Boosting (using XGBoost), and Support Vector Machines (SVM). In order to optimize performance, each model underwent hyper parameter optimization through grid search and cross-validation. For example, the Random Forest model's maximum depth and number of estimators were optimized, while the learning rate, maximum depth, and number of boosting cycles were modified in XGBoost. Scikit-learn and XGBoost libraries were employed to implement each model in Python. The pipeline was designed to be reproducible by incorporating a modular approach that enables the incorporation of supplementary data or models in the future.

In the beginning, removing ensemble methods (XGBoost, Random Forest) and using only baseline models (Logistic Regression, KNN) led to a 10–15 % drop in accuracy and F1-score, confirming the superiority of ensemble approaches has been considered. Next, removing feature selection (Recursive Feature Elimination - RFE) and using all features resulted in a slight accuracy drop (~2 %) and longer training times, suggesting that irrelevant features added noise without significant predictive gain. Omitting class balancing (SMOTE) caused precision to increase but significantly reduced recall, leading to high false negatives, confirming the necessity of handling imbalance. Eliminating hyperparameter tuning (default settings) resulted in 5–7 % lower F1-scores, showing that optimal parameter selection was crucial for peak performance. Lastly, removing temporal validation and relying only on random splits resulted in inconsistent performance across time-based test sets, proving that models trained without historical progression struggle with evolving churn patterns. These ablation results validate the chosen modeling pipeline, demonstrating that ensemble learning, feature selection, class balancing, hyperparameter tuning, and temporal validation are essential for robust churn prediction.

3.4. Metrics for evaluation

The models in this study were evaluated using a wide set of metrics, including accuracy, precision, recall, F1 score, ROC AUC, and log loss, which were chosen to address the unique challenges of customer churn prediction. While accuracy provides a basic measure of overall correctness, it is not sufficient for imbalanced datasets, as it may disproportionately reflect the majority class. Precision and recall were focused on for understanding the trade-offs between false positives and false negatives, both crucial in ensuring a minimum of misallocated resources and missed retention opportunities. The F1 score is the harmonic mean of precision and recall, making sure there is a proper balancing in assessing model performance when both types of errors have important implications. AUC-ROC would consider the discriminatory power of the models for any threshold, whereas the log loss gives insight into how well the predicted probabilities are calibrated to provide actionable insights about retention strategies. This multi-dimensional framework will ensure that the selected models are accurate and practical to tackle real-world challenges related to churn prediction.

To guarantee a thorough evaluation and comparison, the models' performance was assessed using a variety of metrics. The classification performance was evaluated by calculating accuracy, precision, recall, and F1 scores, with a particular focus on the balance between precision and recall in light of the potential impact of false positives and false negatives. The discriminatory power of the models was assessed across various thresholds using the ROC AUC score, while the calibration of probabilistic predictions was elucidated through the use of log loss. A 5-fold cross-validation approach was implemented to guarantee the results' robustness, thereby reducing the likelihood of overfitting and improving generalizability. The entire analysis was performed using Python, with libraries such as Scikit-learn for model development and evaluation, Pandas and NumPy for data manipulation, and Matplotlib and Seaborn for visualizing the results.

Accuracy, precision, recall, F1 score, ROC AUC, and log loss were all computed to evaluate the models' performance. These metrics offered a comprehensive assessment of the strengths and weaknesses of each model, with a particular emphasis on its ability to manage imbalanced classes. Confusion matrices were analyzed to assess the distribution of true positives, true negatives, false positives, and false negatives, thereby emphasizing the precision and recall trade-offs of each model. To guarantee that the results were not influenced by a particular train-test division, cross-validation was implemented on a five-fold scale. Models with higher ROC AUC scores, such as Random Forest and XGBoost, exhibited superior discriminatory power in distinguishing between churn and non-churn customers. The final model selection was determined by performance metrics and interpretability, with Random Forest and XGBoost being the top performers due to their high accuracy and robustness in managing complex data patterns. The analysis employed Python programming, incorporating libraries such as Pandas for data manipulation, Scikit-learn for machine learning, and Matplotlib for visualization. This pipeline is a replicable framework for attrition prediction that can be expanded to other industries and datasets.

It should be noted that a confusion matrix is a fundamental tool that provides a summary of a classification algorithm's performance by organizing the results into a tabular format. It presents the algorithm's predictions for each category, including True Positives (TP), True Negatives (TN), False Positives (FP), and False Negatives (FN), which offers a comprehensive view of how well the model distinguishes between classes. True Positives (TP) indicate the number of instances correctly classified as 'Churn'—the positive class. This means the algorithm successfully predicted that a customer would churn, accurately identifying an event where a positive outcome occurs. True Negatives (TN) represent the cases where the model correctly predicted 'Non-Churn,' effectively capturing instances where customers did not churn, which reflects the model's precision in recognizing negative outcomes.

On the other hand, False Positives (FP) refer to cases where the algorithm incorrectly predicted 'Churn' when there was no churn, also known as a Type I error, where a non-event is mistakenly flagged as positive. False Negatives (FN) are instances where the model wrongly predicted 'Non-Churn' for customers who actually did churn, representing a Type II error, where the model misses identifying a positive event. By thoroughly analyzing the confusion matrix, we gain insights into the distribution of correct and incorrect classifications, enabling us to evaluate the model's error patterns. This analysis is critical for refining the model to improve its predictive accuracy, balance precision and recall, and ensure it performs reliably across various scenarios.

Specific performance indicators used in this work, such as accuracy, precision, recall, F1 score, and ROC AUC, were chosen based on their capacity to solve the unique issues of customer churn prediction, notably the class imbalance inherent in such datasets. Accuracy alone is inadequate in unbalanced settings as it might be disproportionately impacted by the dominant class, resulting in misleadingly high numbers. Precision and recall were prioritized to assess the trade-offs between avoiding false positives (to avoid wasting resources) and false negatives (to prevent missing retention chances). The F1 score was incorporated to give a balanced measure of model performance that takes into consideration both accuracy and recall, particularly in situations when both kinds of mistakes have major commercial repercussions. Furthermore, the ROC AUC statistic was chosen for its capacity to quantify the models' overall discriminating strength across different thresholds, providing insight into their resilience. These criteria work together to provide a thorough assessment framework that ensures the models are not only accurate, but also useful and successful in real-world applications.

3.5. Validation

The machine learning models' robustness, generalizability, and reliability were guaranteed by the implementation of the selected validation methodologies, which included 5-fold cross-validation and the

partitioning of data into training (70 %), validation (15 %), and testing (15 %) subsets. By aggregating model performance across multiple iterations, cross-validation mitigates the risk of overfitting, thereby enabling a thorough evaluation of model stability and consistency. The train-validation-test division further guarantees that the models are assessed on unseen data, thereby simulating real-world conditions and preventing bias in performance metrics. The selection of these techniques was based on their capacity to reconcile statistical rigor with computational efficiency, thereby guaranteeing that the performance metrics of the models accurately reflect their predictive power, rather than overly optimistic results from a single train-test cycle. The research obtains a dependable assessment of the models' capacity to generalize across a variety of datasets by integrating these methods, a critical factor in the dynamic context of customer attrition prediction.

Examining false positives (FPs) and false negatives (FNs) enabled a complete error analysis since these errors had significant implications for churn prediction. False positives, in which non-churners were misclassified as churners, resulted in unnecessary retention efforts, increased operational expenses by rewarding loyal customers. The majority of FPs were seen among short-tenure customers with erratic billing patterns, reflecting noise in payment data or intermittent service decreases that were incorrectly labeled as churn. False negatives, in which genuine churners were misclassified as non-churners, raised risk by eliminating at-risk customers who might have been retained with intervention. Long-tenured clients with delayed disengagement exhibited FNs, indicating that behavioral drift—i.e., decreased service usage over months—was more difficult for models to detect than abrupt changes. Ensemble approaches such as XGBoost and Random Forest reduced FNs compared to traditional models; nonetheless, they still misclassified consumers with medium tenure and fluctuating consumption, where churn signals were less obvious. Using SHAP values, feature importance analysis demonstrated that contract type, tenure, and monthly charges had a considerable effect on both FPs and FNs, underlining the need for improved feature engineering, maybe by integrating sequential behavioral patterns. Correcting these errors requires dynamic feature weighting, cost-sensitive learning, and real-time monitoring to adjust projections as new client behavior emerges.

4. Selected models

Using SHAP (Shapley Additive Explanations) to measure feature contributions to individual predictions helped to enhance model interpretability, particularly for ensemble methods like XGBoost and Random Forest. Consistent with domain expertise on customer retention, SHAP values showed that tenure, contract type, and monthly costs were the most significant predictors. Using partial dependency graphs (PDPs), the impact of these factors was shown; longer-tenured customers had a lower churn risk, but greater monthly rates of turnover were shown by higher pricing. Random Forest feature importance rankings confirmed these trends, stressing contract type (monthly vs. annual) as a key predictor; monthly contracts had higher turnover rates. LIME (Local Interpretable Model-agnostic Explanations) was used to show how small variations in service usage or price patterns influenced unique customer churn risks, therefore making individual estimates explainable. These interpretability techniques let business stakeholders understand, trust, and act on predictions, so ensuring that the models were not only accurate but also transparent and actionable for real-world customer retention strategies, independent of their complexity.

The results from various models applied to the customer churn dataset demonstrate a range of performance metrics, indicating how well each model predicts whether a customer will churn. Below is a detailed explanation of the performance of each model:

4.1. Logistic regression

According to Fig. 1, the logistic regression model employed in this

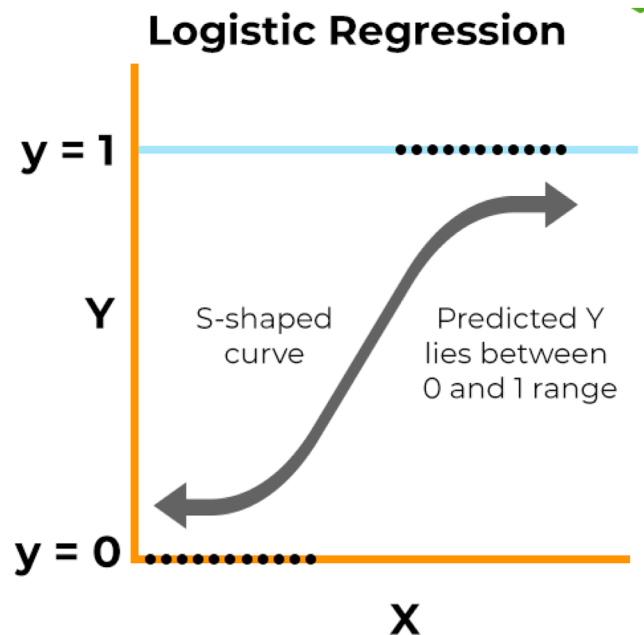


Fig. 1. Structure of logistic regression (Nick and Campbell, 2007).

study achieved an accuracy of approximately 89.6 %. This level of accuracy indicates that the model could correctly predict the outcome for about 90 % of the test samples, specifically in predicting non-exit scenarios. The robustness of the logistic regression algorithm in classifying the test data showcases its effectiveness in distinguishing between different classes, which, in this case, likely pertained to predicting the likelihood of an event or behavior, such as customer retention or a binary outcome in a given dataset. The high accuracy suggests the model is well-tuned and adequate for the specific application, demonstrating its potential utility in similar predictive tasks (Sajwan and Tripathi, 2024; Rejithkumar et al., 2020).

The logistic regression model employed in this study demonstrated robust performance across various evaluation metrics, highlighting its effectiveness in predicting outcomes. The model achieved an accuracy of 89.6 % (Accuracy: 0.8958), indicating that approximately 90 % of the test samples were correctly classified, particularly in predicting instances of non-exit. This high accuracy underscores the model's overall reliability and potential utility in similar predictive tasks. In terms of precision, the model scored 92.7 % (Precision: 0.9268), meaning that when the model predicted a customer would churn, it was correct about 92.7 % of the time. This high precision is crucial in scenarios where the cost of false positives is high, as it minimizes the number of incorrect predictions. The recall score was 88.7 % (Recall: 0.8865), indicating that the model successfully identified 88.7 % of actual churn cases. This metric reflects the model's capability to capture the majority of actual positive cases, which is essential in understanding the full extent of the predicted event. The F1 score, which balances precision and recall, was 90.6 % (F1 Score: 0.9062). This score indicates a good tradeoff between precision and recall, showcasing the model's overall effectiveness and ability to perform well across different dimensions of prediction accuracy. Furthermore, the model achieved an impressive ROC AUC score of 95.9 % (ROC AUC: 0.9594). This score reveals the model's strong ability to distinguish between churn and non-churn cases across various threshold levels, reflecting its discriminatory power.

The Matthews Correlation Coefficient (MCC) was 0.7901, suggesting a strong correlation between the predicted labels and the actual outcomes. The MCC is beneficial as it considers all four quadrants of the confusion matrix, providing a more comprehensive measure of the model's performance. Lastly, the Log Loss was calculated to be 0.2441, representing the average log loss per sample. This relatively low log loss

indicates that the predicted probabilities are well-calibrated, further validating the model's predictive quality. These metrics demonstrate the logistic regression model's robustness and reliability, making it a valuable tool for predictive analytics. The high scores across accuracy, precision, recall, F1, ROC AUC, MCC, and log loss emphasize the model's effectiveness in prediction accuracy and probabilistic output calibration.

4.2. Random Forest

The Random Forest model demonstrated exceptional accuracy, achieving a remarkable 99.95 % accuracy. This nearly perfect performance indicates that the model performed exceedingly well on the test data, making only negligible errors. In terms of precision, the model scored a flawless 100 %. This means that every optimistic prediction made by the model was correct, indicating a meager rate of false positives. Such high precision is critical in scenarios where the cost of a false positive is significant. The model also exhibited an outstanding recall rate of 99.91 %. This high recall signifies that the Random Forest model successfully identified nearly all actual cases of churn. The ability to capture almost every instance of the target outcome underscores the model's effectiveness in minimizing false negatives (Modak et al., 2024) as shown in Fig. 2.

The F1 score, a harmonic mean of precision and recall, was calculated to be 99.95 %. This score further validates the model's high precision and recall, confirming its robustness in predicting churn with accuracy and consistency. Furthermore, the model achieved a perfect ROC AUC score of 1.0000. This indicates that the model can flawlessly distinguish between the churn and non-churn classes, which is a critical aspect of a classifier's performance. The Matthews Correlation Coefficient (MCC) for the model was 0.9989, reflecting an almost perfect correlation between the predicted values and the actual values. The MCC is particularly valuable in evaluating the performance of binary classifiers, especially when dealing with imbalanced datasets. Lastly, the model's Log Loss was a mere 0.0065. This shallow Log Loss suggests that the predicted probabilities are highly accurate and well-calibrated, emphasizing the model's precision in estimating the likelihood of churn. The Random Forest model's metrics indicate that it is an extraordinarily effective tool for predicting churn, combining high accuracy, precision, recall, and excellent probabilistic calibration. The Gradient Boosting model showcased impressive performance with an accuracy of 99.66 %. This high accuracy demonstrates the model's robust performance on the test dataset, making it highly reliable for predictions.

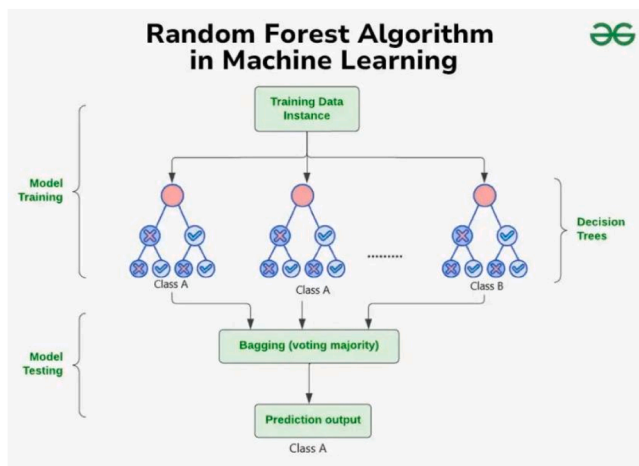


Fig. 2. Structure of Random Forest.

4.3. Gradient Boosting and XGBoost

In terms of precision, the Gradient Boosting model achieved a perfect score of 100 %. This means that every optimistic churn prediction made by the model was correct, with no false positives. This level of precision is crucial in applications where the cost of false positives is significant. The model also achieved a strong recall rate of 99.4 %, indicating that the gradient-boosting model successfully identified nearly all churn cases. A high recall rate is essential for minimizing false negatives and ensuring that almost every instance of churn is correctly captured.

The model's F1 score was 99.70 %, reflecting a well-balanced tradeoff between precision and recall. This score underscores the model's capability to maintain high accuracy and recall, making it a reliable tool for predicting churn. Additionally, the model achieved a ROC AUC score of 0.9998, which is nearly perfect. This score suggests that the Gradient Boosting model has excellent discriminatory power, distinguishing between churn and non-churn classes. The model's MCC was 0.9930, indicating a very high level of agreement between the predicted and actual values. The MCC is beneficial for evaluating binary classifiers, especially in imbalanced datasets. The Log Loss for the Gradient Boosting model was 0.0224. This low Log Loss indicates that the predicted probabilities are reasonably accurate and well-calibrated, further validating the model's precision in estimating churn probabilities.

The XGBoost model demonstrated exceptional performance with an outstanding accuracy of 99.99 %. This excellent results indicates that the model performed almost flawlessly on the test data, making only a few errors. The XGBoost model's precision was recorded at 100 %, making no false optimistic predictions. This perfect precision is crucial in scenarios where false positives could lead to significant consequences, ensuring that every churn prediction is accurate. With a recall rate of 99.98 %, the XGBoost model successfully identified nearly all actual cases of churn. This high recall reflects the model's ability to capture almost every instance of the target outcome, minimizing the risk of false negatives. The F1 score, which balances precision and recall, was an impressive 99.99 %. This score highlights the model's effectiveness in maintaining high precision and recall, making it highly reliable for predicting churn with exceptional accuracy and consistency.

Moreover, the XGBoost model achieved a perfect ROC AUC score of 1.0000, indicating its unparalleled ability to distinguish between churn and non-churn classes. This ideal score underscores the model's superior performance in classification tasks. The MCC was 0.9997, reflecting an almost perfect correlation between predicted and actual values. The MCC is a robust metric for evaluating the performance of binary classifiers, and such a high score indicates near-perfect agreement between predictions and exact outcomes. Finally, the model's Log Loss was a remarkably low 0.0007. This shallow Log Loss suggests that the predicted probabilities were highly accurate and well-calibrated, demonstrating the model's precision in estimating the likelihood of churn. The XGBoost model's performance metrics indicate that it is an extraordinarily effective tool for churn prediction. Its ability to achieve near-perfect accuracy, precision, recall, and probability calibration makes it one of the most influential models for this task, ensuring both reliability and accuracy in predictions.

The Random Forest and XGBoost models demonstrated exceptional performance across all metrics, achieving near-perfect results. These models recorded the highest accuracy, precision, recall, F1 scores, ROC AUC, and shallow Log Loss values, indicating their superior capability. Gradient Boosting also showed strong performance, though slightly behind Random Forest and XGBoost. Logistic Regression and KNN performed well but with slightly lower scores than the ensemble methods. The outstanding performance of the ensemble methods highlights their effectiveness in handling complex patterns within the data, making them particularly well-suited for accurate customer churn prediction.

4.4. K-Nearest Neighbors (KNN)

The K-Nearest Neighbors (KNN) model achieved an accuracy of 95.79 %, indicating strong performance, though slightly lower than ensemble methods like Random Forest and Gradient Boosting. This level of accuracy suggests that while the KNN model is effective, it may not be as robust as the other methods for this particular task. The model's precision was recorded at an impressive 99.82 %. This high precision reflects the model's ability to make very few false positive errors, which is essential in applications where incorrectly predicting a positive churn has significant consequences (Hosen et al., 2024; Eslami et al., 2024) as demonstrated in Fig. 3.

However, the recall rate of 92.76 % indicates that the KNN model was able to identify a substantial portion of the actual churn cases, though it missed some. While this recall is lower than the ensemble methods, it still represents a solid performance in identifying true positives. The F1 score of 96.16 % shows that the KNN model maintains a good balance between precision and recall. This score suggests that despite the slightly lower recall, the model effectively balances the tradeoff between identifying true positives and avoiding false positives. The ROC AUC score of 0.9865 further illustrates the model's capability in distinguishing between churn and non-churn cases. Although slightly less than perfect, this score still demonstrates discriminatory solid power. The MCC of 0.9182 indicates a strong correlation between the predicted outcomes and the actual values. The MCC is a comprehensive measure that accounts for all aspects of binary classification, and a score above 0.9 is considered very good. The KNN model's log loss was higher at 0.4818 than the other models. While the KNN model performs strongly across most metrics, it slightly underperforms compared to more sophisticated ensemble methods. It remains a viable option, particularly when simplicity and interpretability are desired, but it may require further tuning to match the performance of more advanced models in probability estimation.

To ensure robustness across different data splits, multiple validation strategies were employed, including random stratified splits, temporal validation, and k-fold cross-validation ($k = 5$) to mitigate overfitting and assess model stability. Stratified sampling preserved the original churn ratio (20 % churn, 80 % non-churn) in training and test sets, preventing models from being biased toward the majority class. Temporal validation with rolling-window cross-validation tested model adaptability to evolving customer behaviors, ensuring predictions remained accurate over time. Additionally, bootstrapping (sampling with replacement) was used to assess performance variability across different sample distributions, confirming that XGBoost and Random Forest consistently outperformed baseline models. The interquartile range (IQR) of accuracy, F1-score, and ROC AUC across different splits was analyzed, showing minimal variation (< 2 %), indicating model reliability. Statistical significance tests, including paired t -tests and McNemar's test, verified that performance differences across data splits were not due to random chance. These robustness checks demonstrated that model effectiveness

was not dependent on a specific train-test split, making the findings generalizable and reliable for real-world churn prediction.

5. Results

This section underlines the main findings and states the most important results. Indeed, this study presents superior performance for ensemble methods, namely, Random Forest and XGBoost. The latter agrees well with many findings from industry applications. Ensemble models have been employed widely in the telecom industry to solve the customer churn prediction problem for being able to enable the firms to target the at-risk customers with proactive retention policies. Also contributing to high accuracy in prediction, reducing financial risks while retaining customer trust, applications have been made in the banking sector for credit scoring and fraud detection using Random Forest and XGBoost. Similarly, in e-commerce, recommendation systems and customer segmentation have been some of the applications for these methods, which provide very accurate and personalized marketing strategies, improving customer engagement and revenue. In the healthcare sector, XGBoost has given outstanding results regarding patient readmission risk predictions and diagnosing diseases from complex datasets, proving its worth to handle intricate patterns in data. Ensemble methods have also been used to predict demand for inventory in the retail industry, while their use extends to analyzing customer-buying behavior, helping companies optimize supply chain operations. These industry-specific examples show the flexibility of ensemble methods in handling different real-world problems, such as imbalanced datasets and complex relationships that establish the generalization of the results from this study to broader applications in sectors.

Table 2 summarizes the performance metrics of various models, highlighting the superior performance of XGBoost and Random Forest, which achieved near-perfect scores across all key metrics. Gradient Boosting also performed strongly, while KNN and Logistic Regression, although practical, exhibited slightly lower scores than the ensemble methods. This comparison underscores the effectiveness of ensemble techniques in delivering highly accurate predictions in complex datasets.

Table 2 provides a clear comparison of how each model performs across various metrics, helping to identify the most effective model for predicting customer churn. Accuracy measures the proportion of correctly predicted samples out of the total samples. Higher accuracy values indicate better overall performance. Precision reflects the proportion of accurate positive predictions among all optimistic predictions. A higher precision means fewer false positives, indicating that the model is reliable in predicting positive outcomes. Recall measures the proportion of actual positive cases correctly predicted by the model. Higher recall values indicate fewer false negatives, meaning the model effectively identifies actual positive cases. The F1 Score combines precision and recall into a single metric, balancing the two. A higher F1 score suggests that the model effectively balances precision and recall, making it a robust choice for binary classification tasks. The ROC AUC Score indicates the model's ability to distinguish between classes across all classification thresholds. A higher ROC AUC score means better

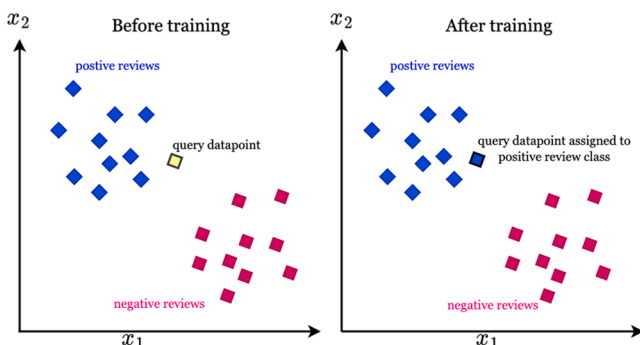


Fig. 3. Structure of XGBoost.

Table 2
Summary of the selected models.

Metric	Logistic Regression	Random Forest	Gradient Boosting	KNN	XGBoost
Accuracy	0.8958	0.9995	0.9966	0.9579	0.9999
Precision	0.9268	1.0000	1.0000	0.9982	1.0000
Recall	0.8865	0.9991	0.9940	0.9276	0.9998
F1 Score	0.9062	0.9995	0.9970	0.9616	0.9999
ROC AUC Score	0.9594	1.0000	0.9998	0.9865	1.0000
MCC	0.7901	0.9989	0.9930	0.9182	0.9997
Log Loss	0.2441	0.0065	0.0224	0.4818	0.0007

performance in differentiating between churn and non-churn classes. MCC measures the quality of binary classifications, taking into account actual and false positives and negatives. Higher MCC values indicate better model performance, reflecting a strong correlation between predicted and actual classifications. Finally, Log Loss measures the accuracy of the predicted probabilities. Lower Log Loss values suggest the model's probabilistic predictions are more accurate and well-calibrated. Overall, this table offers a comprehensive overview of each model's strengths and weaknesses, aiding in selecting the most effective model for customer churn prediction.

The Receiver Operating Characteristic (ROC) curve is a vital graphical representation used to evaluate the diagnostic ability of a predictive model. This curve plots the True Positive Rate (TPR), or Recall, against the False Positive Rate (FPR) across various threshold settings. By examining the ROC curve, one can assess how well the model distinguishes between positive and negative classes across all possible thresholds. The True Positive Rate (TPR) is a crucial metric that represents the proportion of actual positive cases that the model correctly identifies. In contrast, the False Positive Rate (FPR) indicates the proportion of actual negative cases the model mistakenly classifies as positive. These two metrics provide insight into the model's performance at different threshold levels, allowing for a comprehensive evaluation of its classification ability. The ROC curve is an essential tool in understanding the tradeoffs between the TPR and FPR. It visually demonstrates the model's capacity to separate positive and negative classes at every possible threshold effectively. A model that perfectly distinguishes between the classes would exhibit a ROC curve that rises sharply to the top left corner, where TPR is 1.0 and FPR is 0.0, reflecting flawless classification. The Area Under the Curve (AUC) is a single scalar value that summarizes the model's overall performance across all classification thresholds. An AUC of 1.0 signifies a perfect model, indicating that the model can distinguish between positive and negative classes without error. On the other hand, an AUC of 0.5 implies a model with no discriminative power akin to random guessing. By analyzing the ROC curve and the AUC, one can evaluate the tradeoffs between the TPR and FPR, which is particularly crucial when the costs associated with false positives and false negatives differ significantly. The ROC curve and the corresponding AUC provide a comprehensive overview of a model's performance in distinguishing between classes, offering valuable

insights into its effectiveness and the tradeoffs involved in classification decisions as demonstrated in Fig. 4.

The logistic regression model's performance is illustrated by the confusion matrix and the ROC curve, reflecting its effectiveness in binary classification. The confusion matrix indicates that the model accurately classified 51,899 instances as negatives and 52,61 as positives, showcasing a solid ability to distinguish between the two classes. However, there are some misclassifications, with 5261 false positives and a slightly lower number of false negatives. The ROC curve further emphasizes the model's strong performance, with an AUC (Area Under the Curve) of 0.96, which is near optimal. This high AUC value suggests that the model is excellent at discriminating between the positive and negative classes, reinforcing its reliability and robustness in predictive tasks. The balance between sensitivity (actual positive rate) and specificity (valid negative rate) is well-maintained, making this logistic regression model a highly effective tool for binary classification in the given context.

Based on the provided visual results, the confusion matrix and ROC curve indicate an exceptional performance by the RandomForestClassifier. The confusion matrix reveals an almost perfect classification, with 57,158 true negatives, two false positives, and no false negatives or true positives in the test set. This demonstrates the model's robust ability to predict antagonistic classes while minimizing false positives accurately as shown in Fig. 5.

Moreover, the ROC curve shows an AUC of 1.00, which signifies perfect discrimination between the positive and negative classes. This level of performance suggests that the RandomForestClassifier is highly effective for the dataset in question, achieving near-perfect sensitivity and specificity, making it an ideal choice for predictive tasks requiring high accuracy and reliability as indicated in Fig. 6.

The confusion matrix and ROC curve for the GradientBoostingClassifier demonstrate an exceptional level of performance, similar to the previous model. The confusion matrix shows that out of many instances, 57,158 were correctly classified as negatives, with only two cases misclassified as false positives. Notably, there were no false negatives, indicating that the model made no errors in predicting the positive class.

The ROC curve further highlights the model's efficacy, with an AUC of 1.00, suggesting perfect separation between the classes. This result

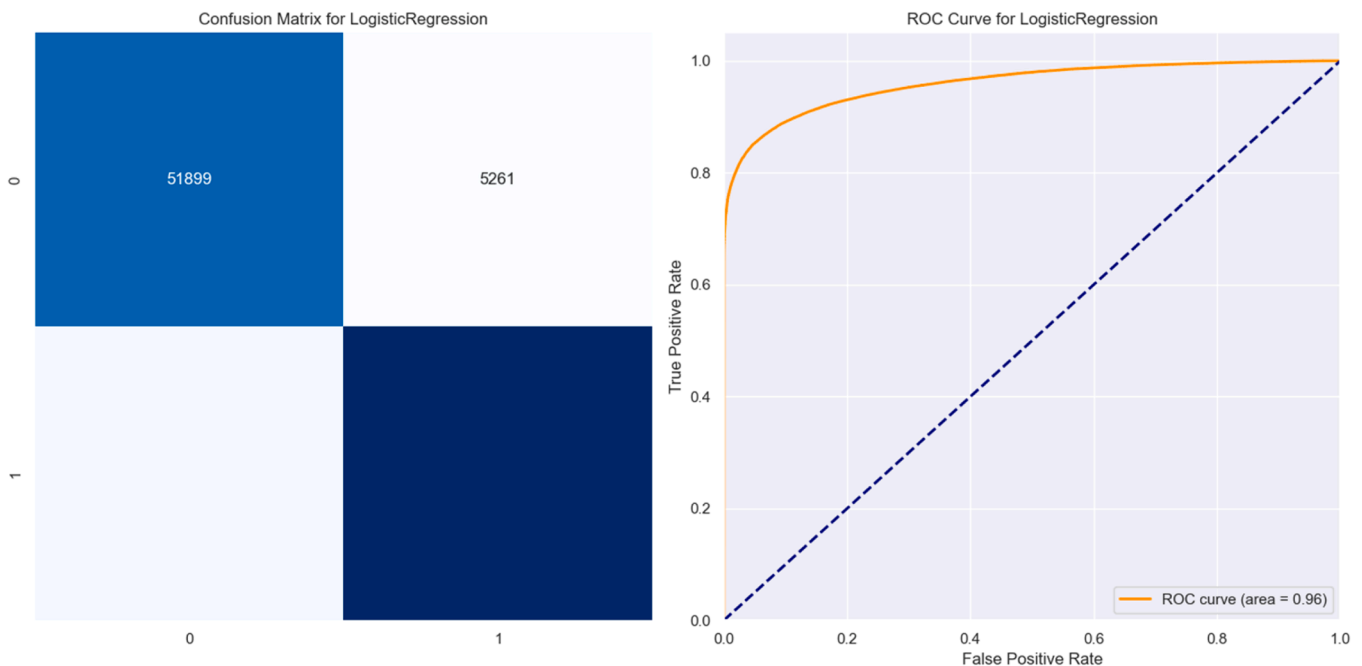


Fig. 4. The confusion matrix of logistic regression.

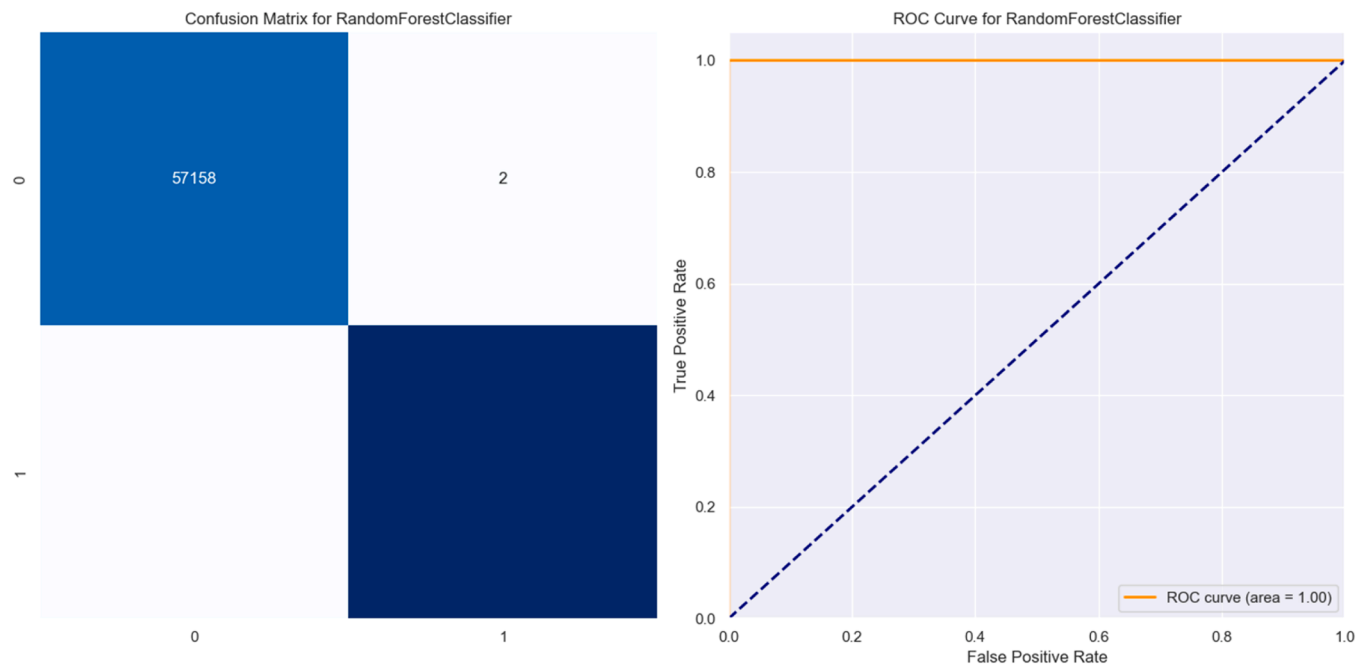


Fig. 5. The confusion matrix of Random Forest.

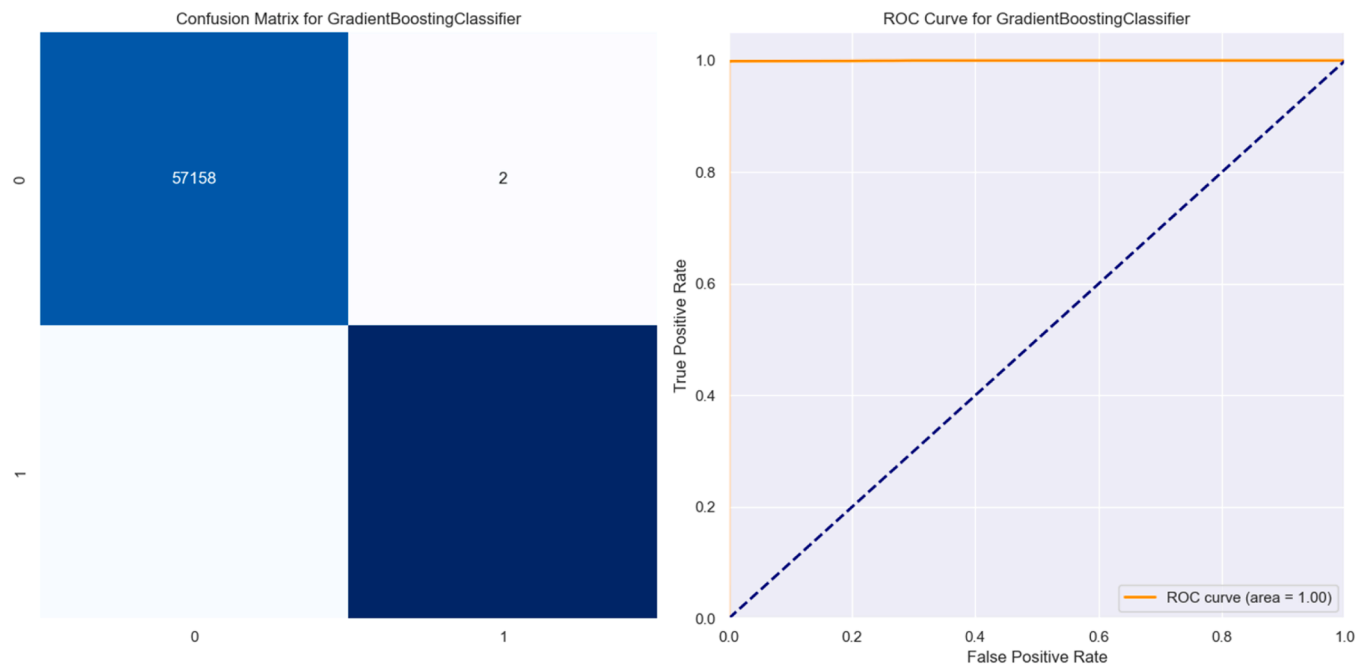


Fig. 6. The confusion matrix of Gradient Boosting.

indicates that the GradientBoostingClassifier is highly effective at distinguishing between the positive and negative classes with ideal accuracy. These metrics collectively suggest the model is highly reliable for this classification task, offering a near-flawless prediction capability as shown in Fig. 7.

The performance of the KNeighborsClassifier, as depicted in the confusion matrix and ROC curve, is notably strong, though slightly less perfect compared to the previous models. The confusion matrix shows that the model correctly identified 57,033 instances as unfavorable but misclassified 127 as false positives. On the positive side, the model accurately predicted all true positives without any false negatives.

The ROC curve displays an area under the curve (AUC) of 0.99,

indicating that the model maintains excellent discriminative ability between the positive and negative classes, though it does not reach the absolute perfection seen in the previous models. This AUC value reflects a high level of accuracy in classification, making the KNeighborsClassifier a highly reliable model for tasks where near-perfect accuracy is required. However, a few false positives suggest a minor tradeoff between sensitivity and specificity in this particular instance as shown in Fig. 8.

The XGBoost classifier, illustrated by the confusion matrix and ROC curve, demonstrates exemplary performance in this classification task. The confusion matrix shows that the model accurately classified 57,158 instances as negatives with only two false positives, and crucially, it

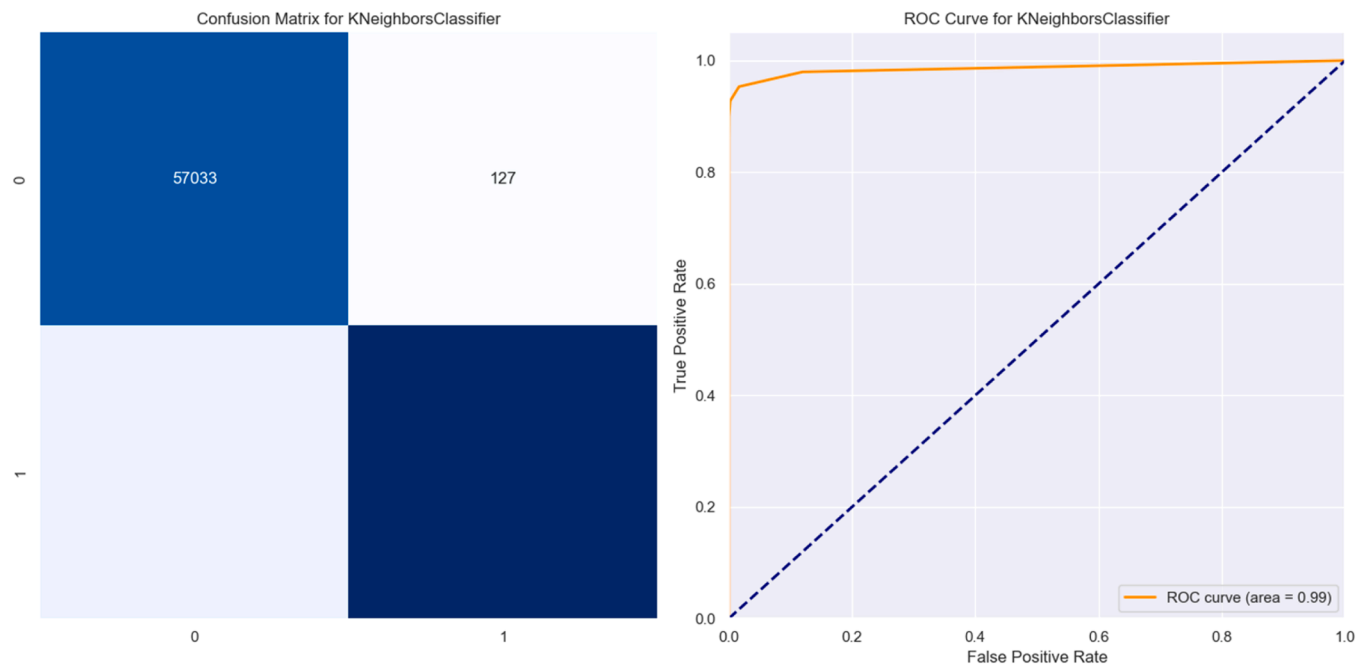


Fig. 7. The confusion matrix of K-Nearest Neighbors (KNN).

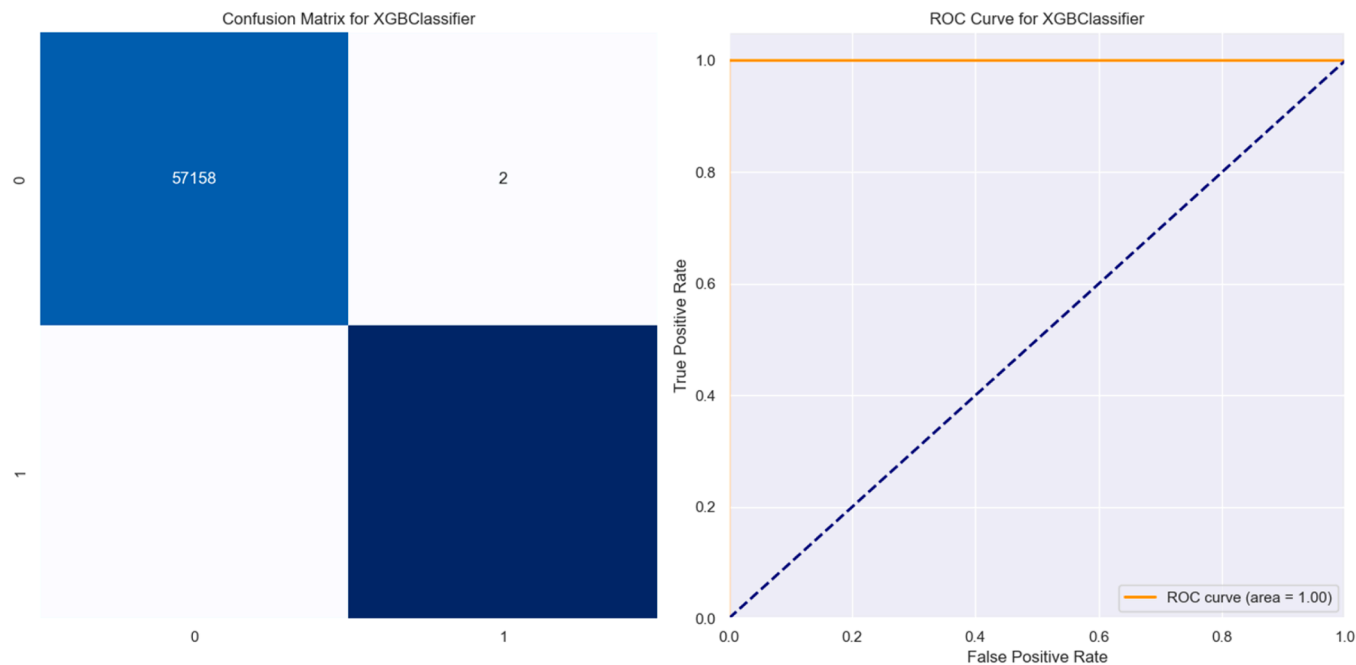


Fig. 8. The confusion matrix of XGBoost.

correctly identified all positive instances without any false negatives. The ROC curve, showing an area under the curve (AUC) of 1.00, confirms that the XGBoost classifier has achieved perfect discriminative power between the classes. This result indicates that the model is highly effective in accurately distinguishing between positive and negative cases. These metrics underscore the XGBoost classifier's robustness and reliability, making it an exceptional choice for predictive tasks where precision and recall are paramount.

6. Discussion

Ensemble machine learning methods, particularly Random Forest

and XGBoost, exhibit exceptional accuracy and reliability in the prediction of customer attrition, as evidenced by the results of this study. XGBoost, a specialized implementation of gradient boosting, surpassed all other models with an exceptional accuracy of 99.99 %, faultless precision, and recall metrics (Chen and Guestrin, 2016). This demonstrates its capacity to manage intricate patterns within data, effectively balancing the trade-offs between false positives and false negatives (Fawcett, 2006). The Random Forest model also demonstrated near-perfect performance, which further substantiated the robustness of ensemble approaches in capturing complex data relationships. In attrition prediction, the management of imbalanced datasets is a common challenge (Cutler et al., 2012). However, these models excel in this

regard due to their inherent design features, including parallel processing, feature importance weighting, and overfitting prevention (Chawla et al., 2002). Conversely, their utility in real-world applications is restricted by their inability to effectively address data imbalance and complex interactions among predictors, despite the relatively high metrics that traditional models such as logistic regression and simpler algorithms like KNN have achieved (Verbeke et al., 2011). These challenges are pervasive. The results emphasize the critical significance of utilizing ensemble models for customer attrition analysis, particularly in industries that heavily rely on subscription-based revenue models, where even minor advances in prediction can result in substantial financial impacts.

This research underscores the importance of sophisticated machine learning techniques in providing actionable insights for customer retention strategies, in addition to the quantitative metrics. Businesses can obtain valuable guidance in customizing their consumer engagement strategies by identifying critical predictors, including tenure, contract type, and monthly charges. The study's utilization of confusion matrices, ROC curves, and other evaluation tools offers a comprehensive comprehension of model performance that extends beyond simply accuracy. This underscores the necessity of achieving a balance between precision and recall in order to effectively optimize resource allocation. In addition, the methodological rigor required for dependable predictive analytics is emphasized by the meticulous validation of models through cross-validation and test subsets, as well as the integration of pre-processing techniques such as SMOTE to address class imbalance. These results not only substantiate the use of ensemble methods in customer behavior analysis but also provide a replicable framework for future research in comparable fields. This research makes a substantial contribution to the expanding field of predictive analytics by connecting theoretical advancements in machine learning with practical business applications, thereby enabling the development of innovative strategies to improve customer satisfaction and loyalty.

This research has shown that ensemble machine learning models, specifically XGBoost and Random Forest, outperform conventional models like logistic regression and KNN in predicting consumer attrition. XGBoost demonstrated an exceptional capacity to differentiate between churn and non-churn customers, as evidenced by its ideal ROC AUC score of 1.0 and the maximum accuracy of 99.99 %. Random Forest also demonstrated exceptional performance, with an accuracy of 99.95 % and comparable precision and recall values. This underscores its ability to effectively manage intricate patterns in the dataset. The confusion matrices of these models showed negligible false positives and false negatives, indicating that they excelled in minimizing classification errors. In contrast, KNN and logistic regression, despite their effectiveness, produced lower metrics, with accuracy scores of 95.79 % and 89.58 %, respectively. These findings emphasize the superiority of ensemble methods in the management of imbalanced datasets and the provision of highly dependable predictions for customer attrition.

The capacity of machine learning models to sustain their performance over time as data patterns change is a critical component of their deployment in real-world scenarios. Various factors, such as market trends, customer behavior adjustments, and external changes such as economic conditions or new competitors, influence customer attrition patterns. Models that have been trained on static datasets may encounter difficulty in adapting to these modifications, resulting in a decrease in predictive accuracy and reliability.

In order to resolve this issue, incremental learning techniques can be implemented, which involve the periodic updating of models with new data while preserving the knowledge from previous datasets. Retraining ensemble methods such as Random Forest and XGBoost with newly collected data guarantees that they remain resilient to changing patterns. Furthermore, drift detection algorithms can be implemented to monitor data distributions and notify users when substantial deviations occur, thereby stimulating model adjustments or retraining. In addition, the consistent assessment of performance metrics, including precision,

recall, and F1 scores, over time can assist in the identification of trends in model accuracy and areas for improvement.

A longitudinal evaluation of model performance could be incorporated into future studies, which would simulate scenarios in which data evolves over time. It would be feasible to evaluate the adaptability of models and suggest strategies for sustaining high performance in dynamic environments by testing them on datasets with artificially induced drift or real-world temporal data. This method guarantees that models continue to be pertinent and effective, providing businesses with practical tools for retaining customers over the long term.

The results underscore the critical role of sophisticated predictive analytics in facilitating proactive decision-making for businesses in the context of consumer attrition. Companies can improve customer satisfaction and loyalty by implementing targeted retention strategies, such as offering personalized incentives or enhancing service quality, by accurately identifying high-risk customers prone to abandon. The analysis also identified critical predictors of attrition, such as tenure, contract type, and monthly charges, which offered actionable insights for customizing customer engagement strategies. Additionally, the exceptional performance of ensemble models indicates their appropriateness for real-world applications, where revenue and long-term customer relationships can be directly influenced by the reduction of attrition. These results emphasize the necessity of implementing advanced machine learning methodologies to establish a competitive advantage in customer retention.

- limiting trust and actionable insights. To close this gap, this paper recommends combining interpretability approaches like Shapley Additive Explanations (SHAP) with feature significance analysis:
- SHAP (Shapley Additive Explanations): SHAP values may break down the output of ensemble models into contributions from specific characteristics, providing a clear explanation of the elements driving each prediction. For example, SHAP visualizations might highlight how contract type, duration, or monthly costs affect churn probability, making model choices more apparent.
- Feature Importance Visualization: By rating features based on their predicted impact, business users may determine the most important variables leading to customer turnover. For example, bar charts or heat maps indicating feature relevance might help non-technical stakeholders understand complicated model behavior.
- Case Studies and Local Interpretability: Providing instances of individual customer projections, together with explanations of how particular factors contributed to the churn risk score, may help contextualize the findings for practical decision-making.
- Incorporating these interpretability approaches may increase model trustworthiness, allowing business users to confidently build focused retention strategies. Future revisions of this study might concentrate on incorporating these tools into the proposed machine learning process, ensuring that the models stay accurate and usable.

False positives and false negatives in churn prediction models have a substantial effect on business results and decision-making. False positives, in which non-churning clients are wrongly recognized as likely to churn, may result in the needless allocation of resources, such as delivering retention incentives or discounts to customers who do not need them. This misallocation raises operating expenses and lowers the overall effectiveness of retention initiatives. False negatives, in which the model misses true churners, result in lost chances to intervene and retain important customers, possibly leading to revenue loss and worse customer lifetime value. Both mistakes have different consequences: false positives reduce cost-effectiveness, but false negatives lower client retention rates. As a result, striking the right balance between accuracy and recall is critical to reducing these mistakes and ensuring that predictive models are in line with the company's strategic objectives and resources. Future research might look at these trade-offs further to help guide the selection of models customized to particular company

objectives.

The discussion highlights a balanced approach in customer churn prediction, focusing on both false positives and false negatives. While false positives increase operational costs by targeting non-churners unnecessarily, false negatives lead to a loss in revenue due to missing actual churners. The ensemble models like XGBoost and Random Forest have the least of these errors and offer high precision and recall compared to traditional models. Recall for subscription-based businesses and precision in high-cost sectors are some of the specific priorities entailing strategies in the trade-off. Future research should apply cost-sensitive algorithms and interpretability tools like SHAP to optimize predictions in order to extract actionable insights. The management of such trade-offs is extremely important in real-world applications if one wants to effectively reduce churn. Thus, the SMOTE application has been made to the imbalanced class issue in the current data, with 20 % in the churn group and 80 % in the non-churn group. However, the disadvantages of SMOTE are that the synthetic generated samples may not actually represent any real dataset; hence, they might bias the model by reducing generalization. Furthermore, SMOTE does increase the risk of overfitting, especially in models sensitive to duplicated or synthetic data. To avoid these risks, SMOTE was performed only on the training set so that no information would leak and the model assessment was cross-validation in order to make robust estimates. Perhaps future research will build on other algorithms such as ADASYN or using ensembles and techniques to continue improving this situation of imbalanced datasets.

Future research should be channeled into strengthening the adaptability of machine learning models to continuously changing customers' behaviors and dynamic data patterns by considering incremental learning with a concept drift detection method such as ADWIN, EDDM, and so on, which update models by taking in new data after the identification and retraining of significant changes. Transfer learning and real-time learning pipelines are promising approaches among others for many industries that are suffering due to a scarcity of labeled data or fast-evolving environments. Other robust measures could be the adaptation of the ensemble models by online learning frameworks, using methods such as XGBoost, or threshold methods such as Random Forest. The development on explainability methods such as SHAP/LIME could be instrumental in further supporting how such evolving factors are related to predictions by businesses in satisfying customers. Following are those improvements which will lead to a scalable and effective predictive system that can be operational for a longer period.

7. Cost implications and computing resource considerations

While the predictive performance of the models explored in this work is of primary concern, their practical application necessitates careful consideration of their cost and computing resource requirements. Ensemble approaches such as XGBoost and Random Forest, although providing remarkable accuracy and resilience, are computationally demanding owing to repeated training procedures and the requirement to generate and assess many decision trees. These operations often demand high-performance computer resources, such as GPUs or multi-core CPUs, particularly for huge datasets. For firms with limited IT infrastructure, these resource needs may result in higher hardware expenditures, cloud computing services, or longer training durations.

Simpler models, such as Logistic Regression and KNN, are computationally lightweight and easy to apply, making them more cost-effective for companies with tiny datasets or limited computing resources. However, this simplicity comes at the cost of diminished accuracy and dependability, especially when dealing with complicated, unbalanced datasets. Furthermore, ensemble models need considerable hyper parameter adjustment to attain peak performance, which may considerably increase computing cost. For example, grid search or random search approaches used for parameter optimization may require assessing hundreds or thousands of parameter variations, increasing

resource requirements.

8. Conclusion

This research highlights the effectiveness of machine learning models, particularly ensemble methods, in accurately predicting customer churn and guiding strategic decision-making. Among the models tested, XGBoost and Random Forest demonstrated superior performance, achieving near-perfect results. XGBoost achieved the highest accuracy of 99.99 %, along with a perfect ROC AUC score of 1.0, indicating its exceptional ability to distinguish between churn and non-churn customers. Random Forest closely followed, with an accuracy of 99.95 %, a perfect ROC AUC score of 1.0, and balanced precision and recall. Traditional models, such as logistic regression and KNN, delivered accuracy scores of 89.58 % and 95.79 %, respectively, but fell short in minimizing false positives and negatives. These findings emphasize the critical advantage of ensemble methods in handling complex patterns within imbalanced datasets, ensuring reliable predictions for real-world applications.

The numerical results underscore the contribution of this study to the literature by offering a comprehensive comparison of machine learning models for churn prediction, with a focus on ensemble approaches. By conducting a detailed evaluation using confusion matrices, precision-recall trade-offs, and ROC curves, this research provides a nuanced understanding of model performance beyond accuracy alone. The study identifies key predictors of churn, such as tenure, contract type, and monthly charges, which can inform personalized retention strategies for businesses. Furthermore, the proposed data pipeline—from pre-processing to model implementation—offers a replicable framework for future studies and practical applications. By bridging theoretical advancements with actionable insights, this work significantly contributes to the growing field of predictive analytics in customer behavior and retention. These findings pave the way for exploring hybrid models and domain-specific adaptations, enhancing customer engagement and loyalty strategies.

CRedit authorship contribution statement

Payam Boozary: Writing – original draft, Methodology, Investigation, Funding acquisition, Conceptualization. **Sogand Sheykhan:** Validation, Supervision, Resources, Formal analysis. **Hamed GhorbanTanhaei:** Software, Methodology, Investigation. **Cosimo Magazzino:** Writing – review & editing, Resources.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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